1st april 2026 - 7th april 2026

Research Project Periodic Log Entry Form

Project ID : R26-SE-039
Student ID : IT22303684
Name : Abeygunasekara D.T
Individual Component : Intelligent Test Case Generation.

Summary of RP work Carried out during the reporting periods.

Date	RP work Conducted
02.04.2026	• Integrated Monaco Editor 0.46 into the gherkin Editor Screen of the React dashboard.
	• Wrote a Custom Monarch tokenizer for gherkin Syntax highighting (Feature, Scenario, Given, when, then)
	• Implement a hybrid Generotor : if the page content match a known template, otherwise call CodeLLM
05.04.2026	• Wired the monco Editor to the Fastapi backend. Editor now loads the latest AI generated Scenario for the selected user story. (GET → api/v1/gherking/{id})
	• Added "Regenerate" button that re runs the hybid jinja2+ CodeLlama generator and a 'Mark as approved' toggle that flips the edited by the @a flag in the ghreakin Scenrio table.

Supervisor's Comments : currently continues the Scope implementations

Student processing well in research Project : (Yes) / NO

if no:

project work to be carried out before next evaluation period : Improve the model accuracies.

Code base checked : Yes / NO

Signature :

Date : 20/04/2026

8th april 2026 - 15th april 2026.

Reasearch Project Periodic Log Entry Form.

Project Id : R26-SE-039

Student Id : IT22303684.

Name : Abeygunasekara D.T

Individual Compoent : Intlligent Test Case Gereration.

Summary of RP worked caried out during the reporting period :

Date.	RP work Conducted.
08.04.2026	• Loaded 25 fresh user Stories produced by Compoent 1 from a real recorded Sprint meeting into the shared PostgreSQL database. • Ran the Full stage1 → stage2 → stage 3 pipeline end to end (intake → specy → parsing → Ghercking generator) and inspected every generated.
11.04.2026	• Measured per stage latency • Generation Success rate and Gherkin syntactic validty using the ghrecking official phrser. • Result : 25/25 stories phrased sucessfully. 24/25 produced valid Ghertin on first attempt. • Prepared a 10 minute supervisor demo covering the entire flow

14.04.2026

① Delivered the Supervisor demo.
· Fixed : two Stories produced
Checking with inconsistent
acter names across Scenarios
(Eg : 'Customer' vs 'user')
· added a normalization step
in parse_story() that
Canonicalizes the acter
across all Scenarios, then
pushed the fix to the main
branch.

Supervisor Comments : Good Improvement

Student presenting well in research project : (Yes) / NO

if NO :

Project work to be carried out before next evaluation
Period : complete the Scope reach to 50%.

Code base checked : Yes / NO Signature : [signature]

① Date : 20/04/2026

29th April 2026 - 6th May 2026

Project Id : R26-SE-039
Student Id : It22303684
Name : Abeygunasekara D.T
Individual Component : Inteligent Test case Generation.

Summery of RP work carried out during the repoting Period

① Date	RP work Conducted
02.05.2026 - 03.05.2026	Implemented Dom avare Crowleing with URL validation and logging. So the generated can see the real elements of the application under test. Added dynamic URL parameters and multi-LLM Support to Code review and Mode Setup. added the Claude Provider alongside the ensitg one. Refactored color schema and styles across Components and cleaned up formating.

04.05.2026 - 05.05.2026	Added the testsuits model and API for storing generated test code with version. Enhanced the mode setup page with URL probing and dom Crawling the dom inspector now remember the last URL, mode and framework (local storage) Made req timeouts ajustable for code genration and Crawling after long crawls timed out. Merged the dom-crawl work to the main branch (PR #4)

Supervisor Comments :

Student processing well in research

Project :- Yes / NO

if NO :

Project work to be carried out before next

encaution Period :

Signature :-

30/05/2026

6th May 2026 - 13th May 2026

Project Id : R26-SE-039
Student Id : IT22303684
Name : Abeygunasekara D.T
Invidual Component : Inteligent Test case generation

Summery of RP work carried out during the reporting period.

① Date	RP work Carried out
06.05.2026	Implemented training of a test risk prediction model and Integrated it into the backend API. Verified the prediction endpoint returns a risk Score for a test case.
08.05.2026	Implemented the Agent Explorer the application in a real browser and Streams live event to the UI, with a Set-of-Marks (SoM) overaly on

Screenshots.
updated presentation materials and adjusted the backend port Configuration.

Supervisor Comments :

Student Processing well in research project.
Yes / No

if NO :-

Project work to be carried out before next evaluation period :-

Signature :
30 / 09 / 2026

22nd June 2026 - 29th 2026

Project Id : R26-SE-039
Student Id : IT22303684
Name : Abeygunsekara D.T
Individual Component : Intelligent Test case Generation.

Summery of RP work carried out during the reporting Period.

Date	RP work Conducted.
22.06.2026	Re-created and ran playwright and Selenium Test suits, and a pytest + playwright suite, for the Saucedemo application.
	Extended test suits management with new endpoints and properties. (Framework, version, Active flag)
28.06.2026	Implement live browser preview with real-time screenshots streaming during exploration. and

execution. Built the github Connection module access token Stored and encrypted. Rest API Connect to repository and Install the workflow. Added return of Github Actions runs and a better execution details UI. Merged the novelty branch (PR#5).

Supervisor Comment :

Student processing well in Research Project :- Yes / No

if NO

Project work to be carried out before Next evaluation Period :

Signature :

30/06/2026

15th July 2026 - 22nd July 2026

Project Id : R26-SE-039
Student Id : IT22303684
Name : Abeygunasekara D.T
Individual Component : Intelligent Test Case Generation.

Summery of RP work Carried out during the reporting Period.

Date	RP work Conducted.
19.07.2026 (part 1)	Ported the component into the merged team frontend. Api clients, types, proxy config, module navigation and the user story intake page. Build the Gherkin Editor (Monaco) with GA edits, approval and regeneration, and added the Invest Criteria view for user stories.

19.07.2026.

(part 2)

Added github connection Settings with tokens validation and workflow install's split test script Generation into its own module with pipline Stages.

Backend: accepts a client supplied project id so auth-service projects and test case projects share one UUID. default port set to 8002 to avoid clashing with the self healing API.

Supervisor Comment :-

Student processing well in Research project : Yes/ NO

if NO :-

Project work to be carried out before next evaluation period.

Signature :- ...

30/07/2026

12th August 2026 - 19th August 2026.

Project Id : R26-SE-039
Student Id : IT22303684
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Individual Component : Intligent Test case Generation.

Summery of RP work carried out during the reporting period

Date	RP work Conducted.
14.08.2026 - 15.08.2026	Merged the end to end test case generation module into the team Frontend (CPR #3). Moved to the Neon cloud database. disable the Statement cache for pgbouncer pooled endpoints and made Startup create any mising table. curate the reseorch Summery and removed unused API/config files.

17.08.2026 - 18.08.2026

Integrate the user Story Service. Import Iteration Stories from Component I into test case generation, keeping the original Story Id so re-import is idemptdent.

Added Auth servic Organisation and Iteration reference columns, Pool pre-ping recycle for Neon idle disconnects, and the failed tests and Selectors Api for the Self healing Component.

Exposed test suits and run Executions to the integration API : removed the unused spacy model to Slim the image, Prepared an azure Service deploy Script.

Supervisor Comments :

Student processing well in research project : (yes) / NO

if NO :

project work to be carried out before next execution period :

Code base checked : Yes / NO Signature :-

Date :- 30/08/2026

No:

Date: / /

19th August 2026 - 26th August 2026.

Project Id : R26-SE-039
Student Id : IT22303684
Name : Abeygunasekara D.T
Individual Component : Intelligent Test Case Generation.

Summery of RP work Carried out during the Reporting period

Date	RP worked.
19.08.2026 - 20.08.2026	Built the project traceblity endpoints C Story → Ghrkin → test → Result) with detailed execution Results added the System Status page to the frontend. Integrated Github with the auth Service improved the cypress Configured dropped the azure Scripts and merged the deploy branchs. Improved constaints and detailed plagarism /saucedemo rules to Suite Generation.

24.08.2026	Implemented a sanitizer that repairs common anti-patterns in generated test suits before they are saved. Linked test runs to their Executions and filtered Scenarios to stop duplicates result appearing.

Supervisor Comments:

Student Processing well in Research project : Yes / No

if NO :

Project work to be carried out before Next Execution Period :

Signature : Lin

30/08/2026

26th Augnest 2026 – 2nd September 2026.

Project Id : R26-SE-039
Student Id : IT22303684
Name : Abeygunsekara D.T
Individual Component : Intligent Test Case Generation.

Summery of RP work carried out during the reporting period.

Date	RP work Conducted.
27.08.2026 – 28.08.2026	Built the endpoint used by the RTM compant : Internal Github credentials a light mode traceibility response. (395 → 55) at pre execution Scoping of Scenario results (PR 10# merged). Deploy to Aws ec2 : Docker ignore before deploy. wrote 16 automated tests.

29.08.2026 - 30.08.2026	Improved Pdf report Generation : running header and footer, refined template and matrix later removed the donut charts for clean layout. Fronted : authentication download , log viewing and image display on the encaution page.

Supervisor Comments :

Student processing well in research project : Yes / No

IF NO :

project work to be carried out Next evaluation period :

Signature : 30/09/2026

7th October 2026 - 14th October 2026

Project Id : R26-SE-039
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Name : Abeygunasekara D.T
Individual Component : Intelligent Test case Generation.

Summery of RP work carried out during the reporting period

Date	RP work.
10.10.2026	Research Pannel ask ~~from~~ for a mechanism to independ-tly validate generated test Cases. design Independent validation gate.
8.10.2026	Fined judge false positives. by adding a Severity Scale and known selected facts to prompts. Ran the test suits and the FE build to confirm everything passes.

Supervisor Comment :

Student processing well in research project : Yes / No

if NO :

Signature :

18/10/2025